

Exam Date & Time: 02-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Parameterized Algorithms [NPT-3205]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Consider a vertex cover problem with a graph $G = (V, E)$ and a parameter k . Analyze how kernelization reduces the size of the graph based on the structure of G . Provide an example of a common kernelization rule used. (3)
 - A)
 - B) Evaluate the application of kernelization in the Feedback Vertex Set problem. How does kernelization improve the feasibility of solving this problem on large graphs? (3)
 - C) Choose a branching strategy in bounded search trees and how it affects the runtime complexity of parameterized algorithms. Why is choosing an efficient branching strategy important? (4)
- 2) Critique the effectiveness of bounded search trees for solving the k -Path problem in terms of both theoretical guarantees and practical limitations. (3)
 - A)
 - B) Explain the advantages and limitations of using Iterative Compression for solving parameterized problems? Use the (Problem: k -Path) as an example to explain. (3)
 - C) Identify how can randomization be incorporated into a parameterized algorithm to solve the k -CLIQUE problem, and what are the possible benefits and drawbacks of using randomized techniques for this problem? (4)
- 3) Analyze how the concept of treewidth influences the efficiency of parameterized algorithms for graph problems like the Vertex Cover problem. Explain the relationship between treewidth and the complexity of the problem. (3)
 - A)
 - B) Evaluate the efficiency of a parameterized algorithm for treewidth computation in terms of its running time complexity. How does parameterization impact the overall algorithm design and performance? (3)
 - C) Consider a parameterized algorithm for solving the Knapsack problem using ILP. Given a knapsack with a weight capacity W , a set of items with weights $w_1, w_2, \dots, w_n$ and profits $p_1, p_2, \dots, p_n$, formulate the ILP that models this problem. Then, solve how you would parameterize this problem with respect to the number of items n and the total weight capacity W . (4)

- 4) Consider the problem of finding the maximum sum subset of a set of integers where no two elements in the subset are adjacent in the original array. Examine a dynamic programming approach to solve this problem efficiently. For the given problem, evaluate the time and space complexity of the dynamic programming solution for finding the maximum sum subset, and discuss the trade-offs between the two complexities when solving large datasets. (3)
- A)
- B) Consider two approaches for finding an important separator in a parameterized version of the Vertex Cut problem: an exhaustive search approach and a parameterized DP approach. Evaluate the feasibility of each approach for large graphs with a small parameter k , and explain which is preferable under these conditions. (3)
- C) Utilize the Fast Fourier Transform (FFT) can be applied as an algebraic technique in parameterized algorithms to speed up convolutions. What is the main advantage of using FFT in solving polynomial-based problems? (4)
- 5) Consider two approaches for solving the Feedback Vertex Set problem in a graph: a traditional branching algorithm and the Cut and Count technique. Analyze the advantages of using the Cut and Count technique in terms of runtime and solution accuracy for large instances with a small parameter k . (3)
- A)
- B) Evaluate the effectiveness of applying matroid theory to the k -Cut problem, where the objective is to partition a graph into k connected components. Discuss how matroid concepts could be beneficial or restrictive in structuring a solution. (3)
- C) Identify how ETH (Exponential Time Hypothesis) influences the understanding of parameterized intractability. How does assuming ETH impact the expected runtimes of $W[1]$ -hard problems? (4)

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